
R8: A Lexical Framework for Approximating Cognitive Manipulation Risk in Text

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Abstract

R8 (Cognitive Risk Analyzer) proposes a structural approach to approximating cognitive manipulation risk in Japanese-language text across 12 theoretically grounded categories. This is an exploratory framework at an early stage of empirical development; the claims made in this paper are limited to what the current implementation can support. The multi-category scoring architecture draws conceptual inspiration—not methodological equivalence—from factor-based psychometric models such as the MMPI.

R8 proposes a lexical approximation framework informed by the qualitative analysis of organizational cognitive failure—as documented in studies of military and institutional collapse (Tobe et al., 1991). The theoretical framework integrates Japanese organizational sociology (Yamamoto, 1977; Nakane, 1967) with Western social psychology (Janis, 1972; Milgram, 1963; Kahneman, 2011).

The system produces a single composite score, the Cognitive Manipulation Index (CMI), defined as an approximation of the aggregate density of linguistic signals associated with manipulation pressure against autonomous thinking. Initial calibration against a 217-document corpus (208 with valid CMI > 0) yields, under the standard detection mode (HIGH $\geq$ 41), a Precision of 97.7%, Recall of 35.5%, and F1 of 52.1 for HIGH-risk binary classification, with an exact match rate of 29.8% and within-one-level agreement of 76.0% across all three levels. Under the high-precision mode (HIGH $\geq$ 60), Precision reaches 100% with Recall of 5.0%. Inter-rater reliability between an LLM-based pilot annotator (Claude Sonnet 4.6 and Claude Opus 4.6; Anthropic, 2026) and the author’s v0.7.1 human_label was Cohen’s κ = 0.211 (Fair, n = 204), confirming that the v0.7.1 annotation criteria require structural reading not currently reproducible by general-purpose language

models; independent human raters remain the basis for Phase 2 reliability assessment. The complete corpus and scoring data are publicly available at <https://github.com/takahiro-oss/r8-cognitive-risk> under CC BY 4.0.

1. Introduction

The proliferation of digital information has intensified the challenge of distinguishing legitimate communication from cognitive manipulation. Existing approaches to this problem—including fact-checking systems, hate speech detection, and misinformation classifiers—focus primarily on the veracity of content. However, cognitive manipulation operates at a structural level that precedes factual falsity. A message can be factually accurate while simultaneously exploiting psychological vulnerabilities through urgency, false authority, emotional amplification, and logical misdirection.

This structural dimension of manipulation has been examined qualitatively in organizational research. Tobe et al. (1991) identified the cognitive patterns that preceded Japan’s institutional collapse in World War II—including the suppression of disconfirming information, over-reliance on authority, and emotionally-driven decision-making. A structurally equivalent phenomenon has been documented in Western organizational contexts. Janis (1972) analyzed the Kennedy administration’s approval of the Bay of Pigs invasion, demonstrating that six critical assumptions underlying the operation—including the belief that Cuban air forces could be neutralized before the landing and that the invasion would trigger spontaneous popular uprising—were adopted without critical scrutiny despite the availability of contradictory intelligence (pp. 32–46). Janis termed this failure mode “groupthink”: the deterioration of mental efficiency, reality testing, and moral judgment resulting from in-group pressures. These patterns were not unique to either cultural context; they recur across corporate, political, and media environments.

R8 (Cognitive Risk Analyzer; the designation “R8” derives from Reiwa 8, the Japanese imperial year of its initial public release) proposes a lexical extension of this qualitative tradition. The system operates across 12 risk categories derived from psychometric and clinical psychology frameworks, producing a single composite score: the Cognitive Manipulation Index (CMI). The present paper positions R8 explicitly as an exploratory baseline instrument rather than a validated measurement tool.

A parallel problem emerges from computational linguistics research on machine reading comprehension. Arai (2018) demonstrated that AI systems process text through statistical pattern matching without achieving genuine semantic understanding. This finding is cautionary rather than validating for R8: it highlights the risk that surface-level pattern matching —whether by AI or by lexical tools —may systematically fail to capture meaning. R8 operates within this same constraint.

The present paper describes the theoretical foundation, implementation, and initial calibration of R8 as an open-source lexical auditing framework. Surface-level lexical analysis, while limited in contextual sensitivity, provides a reproducible and transparent baseline for cognitive risk approximation —one that complements rather than replaces deeper semantic analysis.

R8 is designed as a screening tool that approximates the lexical density of manipulation-compatible signals in text. At the current stage of development, the system cannot determine authorial intent, nor does it empirically demonstrate the cognitive effects of detected vocabulary on readers. These constraints arise from the structural limitations of surface-level lexical analysis, detailed in Section 5. Accordingly, the CMI score functions not as a definitive judgment of manipulative intent, but as an initial flag identifying texts that warrant closer scrutiny.

Within these acknowledged limitations, the utility of R8 rests on three properties. First, transparency and reproducibility: the rule-based, dictionary-matching architecture makes the basis for every score fully explicable and manually verifiable. Second, scalability: large volumes of text can be processed consistently against a fixed set of criteria. Third, complementary function: when combined with human judgment and contextual information, R8 serves as a diagnostic aid rather than a replacement for critical evaluation.

Future development directions include the implementation of structural pattern detection (Phase 2 second-layer architecture), false-negative detection through contrastive corpus construction, and extension to multimodal analysis —including audio and video content, made technically feasible by placing a speech-to-text conversion stage (e.g., Whisper) prior to the R8 pipeline. These developments are positioned as research priorities aimed at approximating intent-level assessment, and are detailed in Section 7.

The cognitive difficulty of individual-level defense against manipulation signals is well documented. Cialdini (1984) proposed a two-question verification procedure for authority signals —“Is this person truly an expert?” and

“How trustworthy is this expert?” —while noting that experts may strategically disclose minor unfavorable information to manufacture an appearance of credibility, after which all subsequent information is accepted uncritically (pp. 371–72). Similarly, regarding scarcity-based pressure, Cialdini notes that the emotional arousal generated by scarcity signals actively impairs the deliberative capacity required for self-protection (p. 425). These observations generalize to the broader class of manipulation-compatible signals that R8 targets: the cognitive conditions that make manipulation effective are precisely those that make individual defense unreliable. R8 is designed to address this structural gap by providing pre-interaction text screening —a function that does not depend on the reader’s real-time deliberative capacity.

2. Theoretical Background

The theoretical framework of R8 integrates previously disconnected research traditions under a unified construct: linguistic signals that target System 1 cognitive processing (Kahneman, 2011). This framing positions R8 not merely as a text analysis tool, but as an operationalization of cross-disciplinary convergence in the study of cognitive manipulation.

The framework draws on six bodies of work. Japanese organizational sociology and cultural psychology (Yamamoto, 1977; Nakane, 1967; Tobe et al., 1991) establish the structural conditions under which implicit cognitive coercion operates. Western social psychology (Janis, 1972; Milgram, 1963, 1974; Cialdini, 1984) independently identifies structurally equivalent phenomena, providing cross-cultural grounding. Cognitive psychology (Kahneman, 2011) provides the processing architecture that unifies these observations: manipulation-compatible linguistic signals function by activating System 1 responses—fast, automatic, and effortless—before deliberative System 2 evaluation can engage. Clinical psychology of manipulation informs the conceptual grounding of several categories, with the psychoanalytic tradition (Symington, 1993) providing one documented account of sub-threshold dependency-induction mechanisms. Psychometric methodology (Hathaway & McKinley, 1943) provides the architectural model for multi-category composite scoring. Computational linguistics (Da San Martino et al., 2019; Pennebaker et al., 2015) situates R8 within existing detection approaches.

The novel contribution of this integration is the identification of a shared

functional mechanism across these traditions: each documents a distinct pathway through which linguistic structure bypasses deliberative evaluation. R8 operationalizes this mechanism as a lexical density approximation. The following sections develop each component of this theoretical foundation in turn.

2.1 From Institutional Failure to Cognitive Manipulation: The “Essence of Failure” Framework

The structural analysis of cognitive failure in organizations has been examined qualitatively in Japanese organizational research. Tobe et al. (1991) identified recurring patterns of institutional collapse in the Japanese military during World War II —patterns that transcend their historical context and recur across corporate, political, and media environments.

Four structural patterns identified by Tobe et al. are directly relevant to R8’s design. First, decision-making was dominated by subjective, wishful thinking rather than objective situational assessment —a pattern R8 operationalizes as Emotional Risk. Second, logical argumentation was systematically displaced by hierarchical authority and face-saving behavior —corresponding to R8’s Authority Risk and Logical Risk categories. Third, organizations demonstrated a consistent refusal to incorporate disconfirming information, suppressing data that contradicted prevailing assumptions —a pattern detectable through R8’s Disclaimer Exploit and Anonymous Authority scores. Fourth, and most critically, organizational decisions were governed not by explicit rules but by an implicit atmospheric consensus.

The recurrence of these four patterns across culturally and historically distinct organizational contexts —documented independently in Western social psychology (Janis, 1972) and cognitive psychology (Kahneman, 2011) —provides the cross-disciplinary foundation developed in Sections 2.4 and 2.5.

2.2 “Kuuki” and the Vertical Society: Structural Foundations of Cognitive Coercion

This implicit atmospheric consensus was theorized by Yamamoto (1977) as “kuuki” (空気) —literally “air” —a form of social pressure that operates below the level of explicit argument, yet determines collective outcomes with greater force than formal reasoning. Kuuki functions as an extra-logical coercive force that paralyzes rational dissent without requiring explicit justifi-

cation.

The structural foundation that sustains *kuuki* is identified by Nakane (1967) in her analysis of Japanese vertical society (*tate-shakai*). In this framework, the credibility of information is determined not by its content but by the hierarchical position of its source. Truth, in effect, is assigned by rank rather than derived through reasoning. R8 operationalizes a proxy for this structural dynamic through its False Authority and Logical Misdirection categories. This vertical structure further generates a sharp cognitive boundary between insider (*uchi*) and outsider (*yoso*) groups, constituting the social substrate from which *EnemyFrame* linguistic signals emerge in Japanese organizational text (Nakane, 1967, pp. 46–47).

Yamamoto illustrates the potency of this force through the case of Japan's wartime naval leadership: experts in possession of statistical data, analytical reports, and formal logical argumentation nonetheless executed operations they collectively recognized as indefensible—and subsequently found themselves unable to articulate any rational account of why they had done so (Yamamoto, 1977, pp. 19–20). This account is significant for R8's theoretical grounding: it is not the absence of evidence or reasoning capacity that produces cognitive failure, but the displacement of both by atmospheric coercion. Statistical analysis, documentary evidence, and logical argument are not merely ignored—they are rendered functionally inert.

Yamamoto further identifies the mechanism through which *kuuki* is generated: not as a spontaneous cultural phenomenon, but as a product of deliberate textual construction. In his analysis of Meiji-era press coverage of the Satsuma Rebellion, Yamamoto documents a reproducible operational pattern—the systematic publication of articles depicting the opposing force as a collective of savage aggressors, juxtaposed against the sacralized representation of the legitimate authority as the embodiment of benevolent order (pp. 49). This binary construction—demonized enemy versus sanctified authority—functions to anchor the reader's emotional orientation before rational evaluation can engage. Yamamoto observes that the fabricated nature of such reporting was detectable to any careful reader, yet the atmospheric effect was achieved regardless (pp. 24, 49).

Yamamoto's core finding—that atmospheric coercion is not a spontaneous cultural phenomenon but a reproducible product of deliberate textual construction—transcends its Japanese organizational context and identifies a manipulation mechanism of cross-cultural applicability directly relevant to CMI detection.

2.3 Cognitive Dependency and System 1 Susceptibility

The susceptibility of individuals to manipulation is grounded in the cognitive architecture described by Kahneman (2011). System 1 thinking —fast, automatic, and effortless —operates as the cognitive default, processing incoming information without deliberative evaluation. Individuals relying predominantly on System 1 processing are susceptible to manipulation through urgency signals and disclaimer exploitation —remaining in a state of passive reception that may prevent the activation of autonomous critical judgment. This dependency is not a personal failing but a structural feature of human cognition: System 1 is the default mode, and deliberative System 2 evaluation requires active effort to engage.

R8 is described metaphorically as a “Cognitive Breakwater” (認知的防波堤): the intent is to identify linguistic patterns that may incline a reader toward cognitive dependency rather than deliberation. The CMI is defined not as a direct measure of manipulation, but as an approximation of the density of linguistic signals associated with such pressure. This sub-threshold account is consistent with clinical observations in the psychoanalytic tradition (e.g., Symington, 1993), which documents analogous dependency-induction mechanisms operating below conscious awareness.

2.4 From “Kuuki” to CMI: A Lexical Approximation of Implicit Coercion

2.4.1 Cross-Cultural Convergence The structural dynamics described by Tobe et al. (1991), Yamamoto (1977), and Nakane (1967) are not culturally unique to Japan. Western social psychology has independently identified structurally similar phenomena.

Janis (1972) documented similar atmospheric consensus dynamics in Western organizational contexts under the term “Groupthink.” Milgram’s (1963) obedience experiments demonstrated the cross-cultural tendency toward authority submission. Kahneman (2011) provided the cognitive architecture that unifies these observations: System 1 thinking —fast, automatic, and effortless —is the cognitive default that manipulative texts may exploit. R8’s CMI can be interpreted as an approximation of aggregate System 1 pressure signals in a text.

2.5 Psychometric Foundations

R8’s multi-category scoring structure draws conceptual inspiration from established psychometric instruments that share a common architectural logic: qualitatively observed psychological phenomena are operationalized as multiple discrete indicators, which are then aggregated into a composite score through weighted combination.

The Minnesota Multiphasic Personality Inventory (MMPI; Hathaway & McKinley, 1943) exemplifies this structure. R8 applies this architectural logic to textual cognitive risk: 12 risk categories function analogously to clinical scales, each capturing a distinct linguistic dimension; the CMI represents the weighted composite. This inspiration is acknowledged explicitly as conceptual only.

2.6 Relationship to Computational Linguistics Approaches

The computational detection of cognitive manipulation and propaganda in text has been addressed through several complementary research traditions.

Da San Martino et al. (2019) proposed a fine-grained propaganda detection framework identifying 18 rhetorical techniques at the fragment level in news articles. Their classification system shares conceptual overlap with R8’s 12-category architecture, though the implementation approaches differ substantially: their neural sequence-labeling model achieves fragment-level annotation, while R8 operates through dictionary-based lexical density at the document level. The interpretability trade-off is explicit: R8 sacrifices coverage for auditability.

The MentalManip dataset (Wang et al., ACL 2024) provides a benchmark for manipulation detection in conversational text, reporting that large language models achieve approximately 65% classification accuracy on manipulation judgment tasks. This figure is notable for R8’s positioning: if state-of-the-art LLM-based approaches achieve 65% on conversational manipulation, the interpretability and domain-specificity advantages of a rule-based system may justify its lower coverage in specialized deployment contexts where stakeholders require explainable outputs.

Pennebaker et al.’s (2015) LIWC framework established that lexical selection patterns carry measurable psychological signals, providing psycholinguistic grounding for density-based scoring approaches. R8 shares this architectural logic while applying it to manipulation-specific categories derived from

clinical psychology and organizational behavior rather than general psychological dimensions.

R8 differs from these approaches in three respects. First, current implementation scope: Japanese-language text, reflecting the initial deployment context rather than a theoretical boundary; the underlying theoretical framework draws on cross-cultural cognitive and social psychology (Kahneman, 2011; Janis, 1972; Milgram, 1963), and extension to other languages is a stated research direction (Section 7). Second, architectural choice: dictionary-based rather than neural, prioritizing interpretability over predictive coverage. Third, theoretical integration: explicit derivation from cross-cultural convergence between Japanese organizational psychology (Tobe et al., 1991; Yamamoto, 1977; Nakane, 1967) and Western social psychology (Janis, 1972; Milgram, 1963; Kahneman, 2011).

Two additional bodies of work provide direct empirical grounding for the manipulation patterns that R8 targets. Ross (2014) identified three structural criteria for cult characterization —absolute authority, coercive persuasion techniques, and exploitative harm —and documented the emergence of online cult dynamics in which manipulation operates exclusively through text, without physical co-presence. This observation grounds R8’s focus on digital text as a self-contained manipulation medium. Ben-Ghiat (2020) analyzed the rhetorical strategies of authoritarian political leaders across historical and contemporary contexts, identifying a recurring operational playbook: sustained disinformation campaigns that overwhelm the audience’s capacity for factual adjudication (corresponding to R8’s PropagandaRisk and EnemyFrame categories), authority signals that bypass deliberative evaluation (AuthorityRisk), and fear-based mobilization that activates threat responses prior to rational processing (FearRisk and AbsolutistWords). The parallel operationalization of cult communication (Ross, 2014) and authoritarian political rhetoric (Ben-Ghiat, 2020) within the same detection framework supports R8’s claim to cross-domain applicability —spanning the organizational, commercial, spiritual, and political genres represented in the calibration corpus.

While research on echo chambers and filter bubbles (Sunstein, 2017) addresses the information environment architecture through which manipulative content reaches audiences, these frameworks operate at a different unit of analysis than R8. R8 targets linguistic signals embedded within individual texts, independently of the distributional context in which those texts are encountered. This distinction defines the boundary of the present study’s

claims.

2.7 Expectancy-Value Theory and the Functional Manipulation of Reader Goals

Eccles and Wigfield (2002) proposed Expectancy-Value Theory (EVT) as a framework for explaining achievement behavior through two psychological constructs: the individual's expectation of success, and the subjective value assigned to a given task. Subjective value is further decomposed into attainment value (personal importance of succeeding), utility value (instrumental relevance to future goals), intrinsic value (enjoyment derived from the task itself), and cost (psychological and material sacrifice required).

EVT was developed in educational psychology to describe how individuals autonomously form motivational beliefs. The present study identifies a structural inversion of this framework in manipulative text: rather than emerging from autonomous appraisal, EVT variables are externally inflated, concealed, or fabricated through lexical means. Utility value is amplified through fear and urgency signals; attainment value is elevated through authority and exclusivity framing; expectancy for success is manufactured through unverifiable statistical claims; and cost is systematically concealed or redefined as evidence of growth.

R8's 12 risk categories can be interpreted as theoretically motivated proxies for EVT variable manipulation. EmotionalRisk and FearRisk are theoretically aligned with utility value inflation; AuthorityRisk and PropagandaRisk with attainment value targeting; LogicalRisk and NakedNumber with expectancy fabrication; and DisclaimerExploit and AbsolutistWords with cost concealment. The category vector profiles observed in Phase 1 calibration — particularly the simultaneous EnemyFrame, PropagandaRisk, and FearRisk activation in HIGH-classified texts, and the LogicalRisk 1.0 pattern in charismatic entrepreneur texts — are directionally consistent with this theoretical mapping, though formal validation against EVT constructs requires independent measurement instruments beyond the scope of the current study. This mapping is proposed as a theoretical foundation for Phase 2 operationalization rather than a claim that current CMI scores measure EVT constructs directly. Critically, intrinsic value manipulation — the use of affirmatively valenced vocabulary (compassion, gratitude, light) to simulate authentic engagement — falls systematically outside the detection scope of the current lexical architecture, constituting the primary source of false negatives documented in Section 5.8.

This mapping does not imply that EVT variable manipulation reflects deliberate authorial design. As documented in Section 5.7, R8 detects lexical patterns that function as EVT variable manipulation—inflating utility value, concealing cost, or fabricating expectancy—independent of whether such manipulation reflects intentional strategy or habitual communication pattern. The distinction between the two cannot be resolved through lexical evidence alone.

The convergence between EVT’s motivational architecture and R8’s empirically derived category structure was reached through independent analytical paths, providing conceptual support for the theoretical coherence of R8’s design. The relationship is presented here as structural correspondence, not as a claim that R8 operationalizes EVT constructs directly.

The motivational foundation underlying EVT variable manipulation is further illuminated by need-to-belong theory. Baumeister and Leary (1995) proposed that the need to belong—to form and maintain stable, positive interpersonal relationships—constitutes a fundamental human motivation with pervasive effects on cognition, emotion, and behavior. Manipulative texts systematically exploit this need through the sequential construction of in-group inclusion and out-group exclusion: establishing similarity and shared identity before introducing enemy framing and exclusivity signals (see Section 3.3, *EnemyFrame* and *PropagandaRisk*). This sequential exploitation corresponds structurally to the EVT category of attainment value manipulation—the manufactured sense that belonging to a specific community is personally essential. R8’s *AuthorityRisk*, *EnemyFrame*, and *EmotionalRisk* categories function as partial detection proxies for this motivational targeting, though the sequential structure of the exploitation pattern falls outside the scope of lexical density scoring and is identified as a Phase 2 design target (Section 7.2).

A related mechanism operates through psychological reactance. Brehm (1966) demonstrated that perceived threats to behavioral freedom generate motivational pressure to restore that freedom—a response that can be exploited by scarcity signals that frame inaction as loss of opportunity. In manipulative text, scarcity signals (time-limited offers, quantity restrictions, exclusive access framing) trigger reactance-based approach motivation, overriding deliberative evaluation. Brock’s (1968) commodity theory formalizes this dynamic: objects become more desirable as their availability decreases, and this effect extends from material goods to information and ideas. Brock further demonstrated that the persuasive power of information

increases when access to that information is itself restricted—an effect amplified when both the content and the source are framed as exclusive. These mechanisms underlie the FearRisk and ClickbaitRisk categories in R8’s architecture, and provide theoretical grounding for the ScarcityRisk category proposed for Phase 2 development (Section 7.4).

Cialdini (1984) synthesizes these mechanisms in a behavioral framework. Authority signals—titles, credentials, institutional affiliations—trigger automatic compliance responses through the same System 1 pathway that Kahneman (2011) identifies as the default cognitive mode. Critically, Cialdini demonstrates that it is not the actual authority of the source that triggers this response, but the perceived authority conveyed by symbols—titles, attire, and automobiles—that have been shown experimentally to elicit compliance even without any legitimate underlying credentials (p. 371). An actor playing a president in a television drama was treated as a genuine political authority and questioned about real presidential policy—a compliance response identical to that generated by actual expertise (p. 349). Across these theoretical traditions, a shared functional pattern emerges at the level of individual cognition: each identifies a distinct mechanism through which externally presented signals—whether framed as authority, scarcity, emotional appeal, or value inflation—can engage System 1 processing (Kahneman, 2011) and reduce the probability of deliberative evaluation. R8’s 12 categories are designed to approximate the lexical density of signals operating through these pathways in Japanese-language text.

3. Methodology

3.1 System Architecture

R8 operates as a pipeline of three sequential processes: text acquisition, lexical analysis, and score aggregation. Input sources include plain text files, PDF documents, and web URLs. All acquired text is passed to a unified analysis function regardless of source type.

3.2 Text Normalization (Phase 1)

Prior to analysis, all input text undergoes normalization to reduce orthographic variation. The normalization pipeline applies Unicode NFKC normalization, followed by katakana-to-hiragana conversion via Unicode code-

point mapping (U+30A1–U+30F3, offset -96).

3.3 Operational Definitions of 12 Risk Categories

The 12 categories are derived from cognitive psychology, social psychology, organizational behavior, and rhetorical analysis frameworks. Each category is operationalized as a lexical density measure: the count of matched dictionary terms divided by text length per 100 characters. Table 2 presents the detection target and primary theoretical basis for each category. Complete operational definitions, including annotation criteria and decision rules, are documented in the publicly available annotation criteria (v0.7.1) at <https://github.com/takahiro-oss/r8-cognitive-risk>.

Table 2. Operational Definitions of 12 Risk Categories

Category	Detection Target	Theoretical Basis
EmotionalRisk	Emotional arousal vocabulary	Kahneman (2011); Cialdini (1984)
FearRisk	Fear/urgency induction	Kahneman (2011); Cialdini (1984)
AuthorityRisk	Legitimate authority exploitation	Milgram (1963); Cialdini (1984)
AnonymousAuthority	Unverifiable authority citation	Nakane (1967); Cialdini (1984)
LogicalRisk	Logical misdirection	Janis (1972); Tobe et al. (1991)
StatisticalRisk	False/decontextualized quantification	Kahneman (2011)
HypeRisk	Exaggeration and unfounded superlatives	Kahneman (2011); Cialdini (1984)
ClickbaitRisk	Curiosity-gap and attention-capture framing	Brehm (1966); Brock (1968)
PropagandaRisk	Us-vs-them framing and disinformation patterns	Janis (1972); Ben-Ghiat (2020)
EnemyFrame	Explicit out-group designation	Nakane (1967); Janis (1972)
AbsolutistWords	Absolute/coercive imperatives	Altemeyer (1981); Ross (2014)
DisclaimerExploit	Responsibility-evasion and consent-bypassing language	Kahneman (2011); Cialdini (1984)

Note on implementation scope: Beyond the 12 theoretical categories presented in Table 2, the implemented system includes three supplementary measures that are not part of the theoretical framework presented in this paper. One auxiliary indicator, Naked Number, detects unsourced specific quantities (e.g., “X-fold” or “X yen” claims without citation) and functions as a binary signal designed for combinatorial interpretation with the 12 primary categories rather than as an independent detection target; it contributes to the CMI with a weight of 0.06. Two peripheral indicators serve applied deployment contexts: Sexual Induction Risk (weight 0.04, included in CMI) and Beauty/Diet Hype Risk (weight 0.00, reference-only, excluded from CMI). Beauty/Diet Hype Risk is implemented to capture gender-targeted manipulation patterns (cosmetic, weight-loss, and relationship-marketing content) that frequently address female audiences in commercial contexts; the weight of 0.00 reflects insufficient corpus calibration data at the present stage rather than a judgment that the signal is unimportant, and future calibration with an expanded corpus including a larger sample of gender-targeted commercial content may justify reassignment of a non-zero weight. These supplementary measures represent implementation extensions for applied deployment contexts.

3.4 Two-Layer Authority Detection

Authority Risk employs a two-layer detection architecture distinguishing pseudo-authority from legitimate authority reference. Pseudo-authority markers are scored directly as risk indicators. Legitimate authority markers are scored conditionally: their risk contribution is modulated by the concurrent Hype Risk density.

3.5 Scoring Algorithm

The Cognitive Manipulation Index (CMI) is computed as a weighted sum of normalized category scores. Category weights reflect the relative contribution of each manipulation dimension to overall cognitive risk, derived through theoretical reasoning and iterative empirical calibration on a small pilot corpus. Statistical Risk carries the highest weight (0.16), reflecting the theoretical manipulative potency of false quantification. CMI ranges from 0 to 100.

Threshold levels for risk classification operate in two modes. The standard mode (default) sets HIGH: $\text{CMI} \geq 41$, MEDIUM: $35 \leq \text{CMI} < 41$, LOW: $\text{CMI} < 35$, optimizing for screening sensitivity (Precision = 97.7%, Recall

= 35.5%, F1 = 52.1). The high-precision mode sets HIGH: $\text{CMI} \geq 60$, MEDIUM: $35 \leq \text{CMI} < 60$, LOW: $\text{CMI} < 35$, optimizing for specificity (Precision = 100.0%, Recall = 5.0%). The threshold is treated as a tunable parameter analogous to temperature in language model inference: the appropriate setting depends on the deployment context and the relative cost of false positives versus false negatives. All corpus statistics reported in this paper use the standard mode (HIGH ≥ 41).

3.6 Implementation and AI Disclosure

R8 is implemented in Python 3.x as two modules: `r8.py` (single-document analysis) and `mass_audit.py` (batch processing). The system requires no machine learning infrastructure, operating entirely through dictionary lookup and regular expression matching. All source code is publicly available under CC BY 4.0 at <https://github.com/takahiro-oss/r8-cognitive-risk>.

R8's development was conducted with support from large language models (Claude, Anthropic; Gemini, Google). AI support was provided for: code development and optimization, translation between Japanese and English, drafting assistance for sections of this manuscript, and literature summarization and synthesis.

The following decisions and analyses were performed solely by the author: theoretical framework design, selection and operationalization of the 12 risk categories, construction of the Japanese-language lexical dictionaries, calibration of category weights, all substantive research direction and interpretive choices, manual annotation of the calibration corpus, and critical assessment of limitations and implications.

The author holds an M.S. in Clinical Psychology (Aichi Gakuin University, 2012) and brings approximately 20 years of direct experience in Japanese educational settings.

4. Initial Calibration and Exploratory Testing

4.1 Study Design

The present calibration exercise is explicitly preliminary in scope. The calibration corpus comprises 217 documents collected through automated URL-based scraping using the `mass_audit.py` pipeline, supplemented by manual

collection via OCR-based screenshot processing for social media content. Of the 217 documents, 9 returned CMI scores of 0.0 due to JavaScript-rendered content inaccessibility or English-language source rejection by the Japanese-language pipeline, yielding 208 documents with valid CMI scores ($\text{CMI} > 0$) — a structural limitation documented in Section 5.4. Of the 9 excluded items, 1 carried `human_label = HIGH`; this item is retained in the corpus as a documented acquisition-failure case but is not counted as a false negative in the Precision/Recall analysis reported in Section 4.3.

Ground truth classification was determined through single-rater expert judgment by the author, drawing on clinical psychology training and 20 years of educational practice. A structured rubric was applied on a provisional basis, drawing on two theoretical frameworks already established in the literature: Ross’s (2014) three-criterion model of coercive communication (absolute authority, coercive persuasion techniques, and exploitative framing) and Janis’s (1972) groupthink symptom profile. Expert judgment was applied at the document level, integrating these theoretical criteria with the author’s clinical psychology training (M.S., Aichi Gakuin University, 2012). Following the initial scanning phase, the annotation criteria were consolidated into version 0.7.1, and the full corpus was re-annotated from a zero baseline under v0.7.1.

4.2 Corpus Description

Genre distribution includes: governmental and institutional documents (LOW baseline), commercial manipulation content (investment schemes, MLM, diet/health products), spiritual and pseudoscientific content, political party media (domestic and international state-affiliated outlets), cult and religious organization content, legal information accounts, financial advisory content, and book-format texts (22 chapters across 4 titles; charismatic entrepreneur and educational critique genres). The book-format subset is analyzed separately in Section 5.8 as a cross-genre replication site for the surface-positive vocabulary false-negative pattern.

4.3 Results

CMI scores were obtained across the full 217-document corpus. Of these, 9 documents returned $\text{CMI} = 0.0$ and are excluded from the following summary statistics; all 9 are retained in the corpus as documented cases (acquisition failure or English-language source rejected by the Japanese-language

pipeline). Summary statistics are reported for the 208 documents with valid CMI scores (CMI > 0).

Table 1. CMI Summary Statistics by Automated Classification Level

Level	n	Mean	SD	Min	Max
HIGH	44	49.0	6.6	41.3	67.6
MEDIUM	37	37.3	1.8	35.0	40.9
LOW	127	21.7	8.5	2.6	34.9
Overall	208	30.3	13.5	2.6	67.6

Median CMI across valid documents: 30.6.

Following the consolidation of annotation criteria into version 0.7.1, the full 217-document corpus was re-annotated by the author from a zero baseline. Under v0.7.1, the human_label distribution is HIGH = 122, MEDIUM = 79, and LOW = 16. Of the 122 HIGH-labeled documents, 1 carried CMI = 0.0 and is excluded from the Precision/Recall analysis (see Section 4.1), yielding an effective HIGH denominator of 121 valid documents. Performance metrics reported below are based on this re-labeled annotation.

Exact match rate (automated level vs. v0.7.1 human_label, valid n = 208): 29.8% (62/208). Agreement within one risk level: 76.0% (158/208). For HIGH-risk binary classification under the standard mode (HIGH $\geq$ 41): Precision = 97.7% (TP = 43, FP = 1), Recall = 35.5% (FN = 78), F1 = 52.1. Under the high-precision mode (HIGH $\geq$ 60): Precision = 100.0% (TP = 6, FP = 0), Recall = 5.0% (FN = 115). The high Precision and modest Recall together indicate that the lexical architecture functions as a high-specificity, low-sensitivity screening filter under v0.7.1: documents flagged HIGH are almost always confirmed as HIGH by human judgment, but a substantial proportion of human-HIGH documents fall below the lexical threshold. The quantitative gap between Precision and Recall provides direct empirical motivation for the Phase 2 structural detection layer described in Section 7.2.

To pilot-test the structural difficulty of the v0.7.1 annotation criteria, an LLM-based pilot annotation experiment was conducted using Claude Sonnet 4.6 and Claude Opus 4.6 (Anthropic, 2026) with the v0.7.1 prompt applied to the full corpus. Of the 217 documents, 13 documents were excluded for file-availability reasons (see Section 4.1), yielding 204 documents with valid LLM annotations. Inter-rater reliability between the LLM annotator

and the author’s v0.7.1 human_label was Cohen’s $\kappa = 0.211$ ($n = 204$), corresponding to “Fair” agreement under the Landis and Koch (1977) benchmark and falling below the Phase 2 acceptance threshold of $\kappa \geq 0.60$. The interpretation of this pilot result is developed in Section 5.3.

4.4 Analysis of Discrepancies

Three categories of discrepancy were identified.

The first concerns text acquisition limitations. Documents returned CMI=0.0 due to JavaScript-rendered dynamic content that BeautifulSoup-based scraping cannot access.

The second concerns scalar underdetection of politically framed content. State-affiliated international outlets and domestic political party media exhibited Enemy Frame and Fear Risk activation at maximum density while returning LOW scalar CMI scores. This reflects a systematic limitation of the current scalar aggregation architecture.

The third concerns satirical mimicry. Satirical content (虛構新聞) that intentionally mimics manipulative news structure registers elevated CMI scores relative to expert LOW classification.

4.5 Implications for Dictionary Calibration

The preliminary results indicate that R8’s current lexical dictionary is sensitive to commercially framed manipulation but requires expansion to capture anecdotal manipulation, spiritually framed authority, and satirical mimicry.

5. Current Limitations

5.1 Surface-Level Lexical Matching

The current implementation relies exclusively on surface-level lexical matching and does not account for contextual disambiguation, linguistic variation, or discourse-level manipulation structures. Three specific consequences follow from this design choice. First, the system cannot distinguish between identical surface forms that carry different semantic functions depending on context. Second, paraphrase and lexical substitution allow manipulation-compatible content to evade detection entirely. Third, the scoring architecture measures lexical density across the entire document, which means

locally concentrated high-severity signals are systematically diluted in long-form texts (see Section 5.10 for empirical demonstration).

The false-positive and false-negative implications of these constraints have been empirically observed in the calibration corpus. The author’s own analytical essay on AI sycophancy (corpus ID: note223) provides a documented case of false-positive risk. Category-level analysis reveals that LogicalRisk (density 1.0), AuthorityRisk (density 1.0), and StatisticalRisk (density 1.0) reached saturation, with EmotionalRisk (0.462), FearRisk (0.74), and HyperRisk (0.74) also activated —patterns structurally attributable to the argumentative and critical register of analytical prose rather than manipulative intent. This demonstrates that the system cannot distinguish between logical argumentation as a rhetorical structure and logical misdirection as a manipulation technique.

These limitations are not resolvable through dictionary expansion alone. The ceiling of lexical-only detection has been empirically mapped through iterative dictionary development from version 10 to version 17.

5.2 Language Scope

The current implementation is Japanese-language specific. All dictionaries, normalization procedures, and validation corpora are constructed for Japanese text, and the theoretical framework draws substantially on Japanese organizational sociology and cultural psychology.

This constraint reflects a deliberate scoping decision rather than a technical limitation. Extension to other languages—including English and other East Asian languages—is a longer-term research direction, contingent on the establishment of validated annotation protocols and comparable theoretical grounding in the target language context.

5.3 Calibration Corpus Limitations

The present calibration corpus is insufficient for definitive statistical inference. The use of the system’s author as the sole human annotator constitutes a significant source of bias. Following the consolidation of annotation criteria into version 0.7.1, the full 217-document corpus was re-annotated by the author from a zero baseline; this re-labeling produced the human_label distribution and Precision/Recall figures reported in Section 4.3.

Furthermore, the corpus includes a critical essay authored by the present

author, examining AI sycophancy (corpus ID: note223). This document produced an automated CMI score of 50.5 (HIGH classification) despite the absence of manipulative intent on the author’s part, and was assigned `human_label = LOW` under v0.7.1 criteria. `note223` constitutes the single false positive ($FP = 1$) in the standard-mode HIGH classification reported in Section 4.3. This case is retained in the corpus as a documented instance of intent-unresolved classification: lexical density alone cannot distinguish critical analysis of manipulative discourse from manipulative discourse itself, since both may reproduce similar vocabulary distributions. The persistence of this single FP across the 217-document corpus represents a categorical limitation of any text-internal scoring architecture, not a calibration deficit addressable through dictionary refinement.

A related structural finding concerns the scope of AuthorityHalo and NormativeInduction signals. These patterns are not limited to texts from unknown or fraudulent actors: texts authored by publicly recognized entrepreneurs and creators with established social credibility also exhibit these patterns (e.g., membership community communications from prominent public figures, MEDIUM-rated in this corpus). This observation reinforces the design principle that R8 detects linguistic structures functioning as cognitive manipulation signals, rather than proxies for sender credibility or intent. High CMI scores therefore neither confirm nor deny fraudulent intent; they indicate the presence of System 1-targeting linguistic patterns regardless of sender status. This distinction has direct implications for the interpretation of MEDIUM-classified texts in deployment contexts: a MEDIUM score for a high-credibility sender may warrant the same scrutiny as a HIGH score for an unknown sender, because the lexical architecture cannot distinguish surface-positive authority framing from benign expert communication.

Phase 2 establishes the formal inter-rater reliability assessment with a minimum of two independent human raters using Cohen’s kappa (κ) as the reliability metric, calculated as:

$$\kappa = (P_o - P_e) / (1 - P_e)$$

where P_o denotes the observed proportion of agreement and P_e denotes the proportion of agreement expected by chance. A minimum threshold of $\kappa \geq 0.60$ is adopted as the criterion for acceptable reliability, consistent with the benchmark established by Landis and Koch (1977), who characterized κ values of 0.41–0.60 as moderate, 0.61–0.80 as substantial, and 0.81–1.00 as almost perfect.

An LLM-based pilot annotation experiment conducted in advance of the formal Phase 2 assessment (reported in Section 4.3) yielded Cohen’s $\kappa = 0.211$ ($n = 204$) between Claude Sonnet 4.6 and Claude Opus 4.6 (Anthropic, 2026) and the author’s v0.7.1 human_label — Fair agreement under the Landis and Koch (1977) benchmark, but below the $\kappa \geq 0.60$ acceptance threshold. This pilot result does not substitute for the formal Phase 2 assessment with independent human raters; rather, it indicates that the v0.7.1 annotation criteria — which require structural reading of the two-stage cognitive induction architecture (Section 5.7) and surface-positive vocabulary manipulation (Section 5.8) — impose a judgment load that current general-purpose language models cannot reproduce. The formal κ assessment remains a Phase 2 requirement to be conducted with independent human raters.

5.4 Digital Media Application Constraints

The application of URL-based lexical analysis to digital media faces a hierarchy of structural constraints. At the technical level, a significant proportion of high-risk web content employs JavaScript rendering that prevents direct HTML extraction. Direct URL scanning consistently underperforms text-converted scanning by a substantial margin (CMI 10.0 vs 44.7 confirmed on identical pages).

Beyond the JavaScript problem, a more fundamental constraint emerges from the structural characteristics of social media content. Short-form posts on platforms such as X (formerly Twitter) present a categorically different challenge. During calibration, five posts (corpus ID: sn229–sn233) were scanned via OCR pipeline. All produced CMI LOW despite containing vocabulary consistent with EnemyFrame and Propaganda categories. Two failure modes were identified: density calculation failure for 140–80 character texts, and OCR-induced character-level misreads breaking dictionary matching.

The most fundamental limitation is epistemological. Fact-based disinformation — claims such as “a public figure’s family member holds a specific nationality or affiliation” — cannot be evaluated through lexical analysis alone. This reflects a categorical boundary between lexical pattern detection and factual verification.

Phase 2 design priorities include dynamic threshold adjustment based on text length (short-text mode) and thread-level aggregation enabling multi-post integration before scoring.

5.5 Dictionary Construction Methodology

The current lexical dictionaries were constructed through theoretical reasoning and expert judgment rather than corpus-based induction. The iterative development process from version 10 to version 17 of the R8 dictionary expanded detection coverage for specific genres (spiritual/pseudoscience content in v11–13, political conspiracy content in v14–15, deletion-threat urgency patterns in v16). However, this expansion process has a structural ceiling: the most sophisticated manipulation content remained in the MEDIUM range regardless of dictionary expansion, because the manipulation operates through vocabulary that does not appear in any risk category.

This iterative expansion process introduces the risk of circular validation. Dictionary expansion is therefore reframed not as a precision improvement strategy but as documentation of the lexical detection ceiling before transition to transformer-based contextual analysis. Version 17 represents the current boundary of what surface-level lexical matching can achieve for the genres represented in the calibration corpus.

5.6 The Gap Between Theoretical Framing and Implemented Method

R8 is an open-source, transparent instrument for approximating the density of manipulation-compatible linguistic signatures; surface-level lexical patterns do not establish manipulative intent or effect. A structural limitation that underlies all others in this paper deserves explicit statement. The theoretical framework presented in Section 2 draws on cognitive psychology (Kahneman, 2011), social psychology (Janis, 1972; Milgram, 1974; Zimbardo, 2007), and Japanese organizational sociology (Nakane, 1967; Yamamoto, 1977) to characterize cognitive manipulation as a multi-level phenomenon. R8 is theoretically positioned as a detection system for linguistic signals targeting System 1 cognitive processing (Kahneman, 2011).

The implemented method —surface-level lexical density analysis —operates at a categorically different level of description. No amount of dictionary expansion will bridge the gap between counting words and measuring cognitive effects.

What the gap does require is explicit restraint in theoretical claims. The paper presents R8 as a lexical screening tool that approximates manipulation-compatible signal density, not as a direct measure of cognitive manipulation. The theoretical framing in Section 2 should be read as motivating the category design and providing interpretive context, not as a claim that CMI

scores measure the theoretical constructs directly. This distinction is particularly salient for the limitations documented in Sections 5.7 through 5.10.

5.7 Authorial Intent and the Limits of Lexical Inference

The present system analyzes the linguistic structure of texts but cannot determine the intentional state of the author reflected in that structure. The high-density occurrence of manipulation-compatible vocabulary may reflect deliberate cognitive manipulation design, the author’s own sincere beliefs expressed through habitual language, or any combination of the two.

Symington (1993) describes cases in which narcissistically structured communicators are themselves unaware of the manipulative architecture of their speech. Festinger’s (1957) cognitive dissonance theory suggests a parallel mechanism: communicators who experience tension between their stated framing and their underlying intent may resolve this dissonance by elaborating the framing itself—inadvertently intensifying the manipulation-compatible signal density of their text. The present study has no means of empirically distinguishing these three types (intentional, unconscious, and mixed), and CMI scores therefore carry no implication about the author’s cognitive state or ethical status.

In the calibration corpus, a legal information note account (corpus ID: note164–note179, 15 valid documents) reproduced the stereotyped sequence of fear induction, authority presentation, logical justification, and action induction across all articles—instantiating what the present study terms a two-stage cognitive induction architecture. This structure may represent intentional design or habitual communication pattern; text analysis alone cannot resolve the distinction.

The CMI distribution across this 15-article series provides quantitative evidence of the structural detection gap. All 15 articles received `human_label = HIGH` under v0.7.1 criteria, reflecting their structurally identical two-stage induction architecture. Automated CMI scores, however, ranged from 20.7 to 61.1—a threefold spread (factor 2.95) within a single-author, single-template corpus—and only 8 of 15 articles (53.3%) reached the HIGH threshold ($\text{CMI} \geq 41$) under the standard mode. The remaining 7 articles distributed across MEDIUM ($n=1$) and LOW ($n=6$) classifications despite reproducing the same fear–authority–justification–action sequence. This within-author CMI variance under structural invariance constitutes the most direct empirical evidence available in the present corpus that

lexical density does not track structural manipulation: the pattern that determines human-judged HIGH classification is invariant across the series, while the pattern that determines automated CMI varies by a factor of 2.95. The phenomenon cannot be attributed to between-author stylistic variation, between-genre vocabulary differences, or between-text-length effects, because all three sources of variation are controlled by the single-author, single-template corpus design. The remaining variation must originate in surface lexical fluctuations that the structural reading required by v0.7.1 (H-2 criterion) explicitly disregards. This observation reinforces the argument developed across Sections 5.6–5.8 that the gap between lexical density and structural manipulation is not a calibration deficit but a categorical limitation of the surface-lexical detection paradigm.

5.8 Surface-Positive Vocabulary and Systematic False Negatives

The most structurally significant detection limitation identified in Phase 1 calibration is the systematic underdetection of texts employing affirmatively valenced vocabulary as a manipulation medium.

Cult-affiliated and spiritual texts in the calibration corpus (corpus ID: web138–web197, relevant subset) were classified as MEDIUM-equivalent by the expert evaluator but produced CMI scores consistently in the LOW range. These texts share a characteristic vocabulary profile: high-density use of terms such as “compassion,” “love,” “light,” “walking together,” and “gratitude” —surface-positive expressions that appear in none of R8’s 12 risk categories.

The author’s own analytical essay (corpus ID: note223) provides a particularly clear illustration. A critical analysis of AI sycophancy produced an automated CMI score of 50.5 (HIGH classification) despite explicit analytic intent. The author cannot themselves determine with certainty whether implicit rhetorical intent is entirely absent, instantiating Section 5.7’s fundamental problem: lexical density alone cannot distinguish critical analysis of manipulative discourse from manipulative discourse itself, because both reproduce structurally similar vocabulary distributions.

Distinguishing authentic positive communication from manipulation-mediated positive vocabulary requires three analytical layers: statistical extraction of co-occurrence patterns between positive vocabulary and action-induction language (PMI analysis); inductive sequence analysis reconstructing the communicative destination from the final paragraph

backward; and contrastive corpus comparison against authentic positive text baselines. These constitute Phase 2 research priorities.

A cross-genre replication of this false-negative pattern was identified in the book-format corpus. Seven chapters of a charismatic entrepreneur text (Yozawa, T., 2013, *Byousoku de ichioku-en kasegu jouken* [The conditions for earning 100 million yen per second], Forest Publishing; corpus ID: book001_ch01–ch07) were analyzed using the same pipeline. All seven chapters received HIGH human_label assignment; automated CMI scores ranged from 18.6 to 45.5, with five of seven chapters classified as LOW. LogicalRisk reached saturation (0.74–1.0) across all chapters, and AbsolutistWords reached ceiling (1.0) in two chapters (ch03, ch07), yet EmotionalRisk remained near zero (0.0–0.139) throughout. This profile is structurally identical to the charismatic entrepreneur texts identified in Phase 1 web corpus calibration (Watanabe corpus, 2010–011): LogicalRisk saturation co-occurring with near-zero EmotionalRisk, producing systematic underdetection despite expert HIGH classification. The cross-corpus replication across a publication span spanning the early 2010s suggests that this false-negative pattern reflects a stable structural feature of charismatic entrepreneur discourse rather than an artifact of individual authorial style. This pattern constitutes a transhistorical reproduction of surface-non-negative vocabulary manipulation —a systematic false-negative class for which lexical dictionary expansion offers no structural remedy.

A structurally distinct false-negative pattern concerns the visual and multi-modal dimensions of authority construction. Cialdini (1984) demonstrated that authority compliance is triggered not by the genuine credentials of a source but by the perceived authority conveyed by symbols —titles, attire, institutional affiliations, and the visual presentation of expertise (p. 371). In digital text environments, these authority symbols are instantiated not through vocabulary but through visual elements: profile photographs conveying professional appearance, display of academic credentials or certifications as images, institutional logos, and the visual design language of high-production-value websites. R8 operates exclusively on extracted text and therefore cannot detect any of these visual authority signals. A text that functions as high-risk authority-based manipulation through visual cues may register near-zero AuthorityRisk and AnonymousAuthority scores if the verbal content is stripped of explicit authority vocabulary. This represents a categorical detection boundary rather than a calibration limitation: no expansion of the lexical dictionary can address signals that are not encoded in text. Phase 2 corpus design will include human annotation of visual au-

thority signal presence as a supplementary dimension to enable systematic documentation of this false-negative class.

5.9 Severity Weighting Limitations

The current scoring architecture treats all matched terms within a category as equally weighted contributors to category density. Terms carrying explicit harm-relevant meaning (e.g., references to criminal extortion or threats of physical injury) are scored at the same per-occurrence weight as terms carrying milder fear-induction valence. A severity-weighted scoring scheme would require independent annotation of qualitative severity at the dictionary entry level, increasing both annotation cost and inter-rater coordination requirements; this trade-off is documented as a Phase 2 design consideration rather than a v1 implementation target.

5.10 Short-Form Text Application Constraints

The density-based scoring architecture was designed for medium-to-long-form texts and exhibits structural detection failure for short-form social media content.

During Phase 1 calibration, five posts related to a child homicide case on X (formerly Twitter) (corpus ID: sn229–sn233) were scanned via an OCR-based pipeline. All five produced CMI LOW despite containing vocabulary consistent with EnemyFrame and Propaganda categories. Two failure modes were identified: density calculation failure for 140–80 character texts, and OCR-induced character-level misreads breaking dictionary matching.

The most fundamental limitation is epistemological. Fact-based disinformation—specific factual claims whose falsity cannot be assessed through lexical analysis—falls categorically outside the detection scope of the current system. This marks a categorical boundary between lexical pattern detection and factual verification, not resolvable through dictionary expansion.

Phase 2 architectural responses include dynamic threshold adjustment calibrated to text length (short-text mode), thread-level aggregation processing multiple posts as a unified document before scoring, and hashtag pattern analysis as a supplementary signal channel.

A further failure mode was identified during Phase 1 corpus construction: two documents (corpus ID: sn234, sn235) were acquired via OCR-based screenshot processing but returned CMI = 0.0 due to capture of non-target

content — a NAS management interface and a Claude session window respectively. These documents are retained in the corpus as documented cases of acquisition failure and flagged for deletion prior to Phase 2 re-labeling. This failure mode illustrates a categorical limitation of screenshot-based corpus construction: OCR cannot distinguish between the target text and surrounding interface elements, and acquisition validity cannot be confirmed without manual inspection of each document. Phase 2 corpus design will require a mandatory content-validity check prior to annotation.

6. Discussion

6.1 R8 as a Screening Tool: Utility and Acknowledged Limitations

R8 is explicitly positioned not as a definitive classifier of manipulative intent, but as an initial flag that complements rather than replaces human judgment. The system’s utility rests on three properties: transparency and reproducibility, scalability, and complementary function.

This positioning is analogous to medical blood testing: blood tests do not definitively diagnose disease but identify abnormal values requiring closer examination. R8 detects “signs requiring scrutiny” in text; it does not determine manipulative intent. Explicit disclosure of this limitation is essential for preventing misuse of the instrument.

The corpus distribution observed in Phase 1 calibration —LOW: 61.1%, MEDIUM/HIGH: 38.9% —should be interpreted within this screening framework rather than as evidence of low detection power. A first-stage screening instrument is designed to efficiently exclude low-risk documents from further analysis while directing resources toward documents requiring closer scrutiny. Of the 217 documents, 9 returned CMI = 0.0 and are excluded from scalar screening analysis, yielding 208 valid cases. R8 identifies 38.9% of documents (81/208 valid cases) as requiring attention—a proportion that, within a screening framework, functions as a first-stage filter directing analytic resources toward a tractable subset. Specifically, documents classified as MEDIUM or HIGH are flagged for closer examination of category vector profiles, co-occurrence patterns, and contextual factors that fall outside the scope of lexical density scoring. The false-positive rate within this flagged subset is low: 1 of 44 HIGH-classified documents received non-HIGH human_label assignment under v0.7.1 (2.3%), indicating that the first-stage filter produces few spurious escalations. However,

the complementary figure demands equal transparency: 115 of 127 LOW-classified documents received MEDIUM or HIGH human_label assignment (90.6%), indicating that the majority of documents cleared by the first-stage filter may warrant re-examination under a second detection layer. The LOW classification does not assert absence of manipulation; it asserts that the lexical signal density does not reach the threshold justifying escalation. The magnitude of the false-negative rate —empirically documented here under the consolidated v0.7.1 annotation criteria —provides direct quantitative motivation for the Phase 2 structural detection architecture described in Section 7.2.

The Recall figure of 35.5% under v0.7.1 reflects the substantive change in the annotation reference rather than a degradation in lexical performance. Under v0.7.1, the human_label HIGH set expanded as the criteria became more inclusive of structurally manipulative texts that operate through surface-positive vocabulary and two-stage cognitive induction —patterns documented in Sections 5.7 and 5.8. Because the lexical architecture cannot detect these patterns, the increase in human-HIGH documents enlarged the false-negative count without corresponding gains in lexical recall. This quantitative shift directly substantiates the surface-positive vocabulary false-negative limitation (Section 5.8) and reinforces the case for Phase 2 structural pattern detection as the necessary next architectural layer.

6.2 CMI Scores and Category Vector Profiles

A key finding of Phase 1 calibration is that the 12-category vector profile is diagnostically superior to the scalar CMI value. Political and conspiracy-theory texts reached HIGH classification through simultaneous elevation of EnemyFrame, Propaganda, and FearRisk —a pattern that, based on Phase 1 calibration data, shows directional convergence with elements of Janis’s (1972) groupthink symptom profile, though the analogy is subject to the structural limitations described below. An empirical observation from Phase 1 calibration provides a partial point of contact: under the high-precision mode ($\text{HIGH} \geq 60$), all six HIGH-classified documents exhibited simultaneous activation of EnemyFrame, FearRisk, and PropagandaRisk—the three categories directionally corresponding to Janis’s stereotyped out-group framing, vulnerability illusion, and pressure toward uniformity. Conversely, charismatic entrepreneur texts that plateaued at MEDIUM showed near-zero EnemyFrame activation, consistent with Janis’s observation that out-group designation accelerates groupthink dynamics. This

co-occurrence pattern is interpreted as directional convergence rather than empirical validation: the sample of six HIGH documents is insufficient for statistical inference, and the correspondence between R8 lexical categories and Janis’s group-process symptoms operates across categorically different levels of analysis. Three structural differences further limit the analogy. First, Janis described closed-group face-to-face decision-making processes, whereas R8 targets one-directional text addressed to an unspecified audience. Second, Janis’s unit of analysis is the group; R8’s unit is the relationship between an individual reader and a text. Third, Janis documented decision-making processes observable by external researchers; R8 detects lexical surface features that may or may not reflect the group-level dynamics Janis theorized. No systematic empirical mapping between R8 categories and Janis’s eight symptoms has been conducted in this study.

Category vector profile analysis can visualize differences invisible to scalar scores. FearRisk elevation in isolation suggests fear-induction manipulation; Authority combined with Hype suggests pseudo-authority exploitation.

6.3 Surface-Positive Vocabulary and the False-Negative Problem

The most important structural finding of Phase 1 calibration is the systematic false-negative pattern for texts using affirmatively valenced vocabulary as a manipulation medium (Section 5.8). The finding is not resolvable through dictionary expansion.

The author’s own analytical essay (corpus ID: note223) illustrates this problem. Critical writing about AI sycophancy produced a CMI HIGH classification (CMI = 50.5), with the author unable to determine with certainty whether implicit rhetorical intent is entirely absent. Phase 2 priority: construction of a contrastive corpus pairing authentic positive texts with manipulation-mediated positive texts to enable false-negative detection.

6.4 Short-Form Social Media Content

Phase 1 calibration confirmed that X (formerly Twitter) posts present a structurally distinct challenge for density-based scoring (Section 5.10). The scoring architecture is not suited for texts of 140–80 characters, and fact-based disinformation falls categorically outside lexical detection scope. Phase 2 architectural design priorities: dynamic threshold adjustment (short-text mode) and thread-level aggregation.

6.5 Circular Validity and the Need for Independent Validation

The most critical methodological constraint of this study is the circular structure in which the system designer serves as sole evaluator. Dictionary design, threshold calibration, and human_label assignment were all performed by the same individual, with no independent external validation.

Phase 2 establishes annotation standard consolidation and a minimum of two independent raters with Cohen’s kappa reliability measurement as necessary conditions before corpus expansion. The LLM-based pilot annotation experiment reported in Section 4.3 (Cohen’s $\kappa = 0.211$, $n = 204$) confirms that this independence requirement cannot be circumvented by LLM-based annotation: current general-purpose language models cannot reproduce the structural reading required by v0.7.1, and therefore cannot substitute for human raters in the Phase 2 reliability assessment. Independent human annotation by a minimum of two raters remains a non-negotiable prerequisite before Phase 2 corpus expansion proceeds.

7. Future Work

The development roadmap is structured in two phases. Phase 2 addresses the empirical foundations required before the current framework can support stronger claims. Phase 3 work is not planned to begin until Phase 2 has produced an independently annotated corpus of at least 200 documents with documented inter-rater reliability (Cohen’s kappa ≥ 0.6) and a CMI-to-annotation agreement rate of at least 65% exact match on the independent corpus.

7.1 Phase 2: Validation Corpus Expansion and Inter-Rater Reliability

Expansion of the calibration corpus and establishment of inter-rater reliability are the immediate empirical priorities. The Phase 2 corpus design will address current limitations through: (1) manual collection of HIGH-risk documents where automated collection fails due to JavaScript rendering; (2) addition of a contrastive corpus of authentic positive texts (administrative documents, medical informed consent forms); (3) inclusion of social media content (SNS); (4) independent annotation by a minimum of two raters with Cohen’s kappa measurement ($\kappa \geq 0.60$; Landis & Koch, 1977). The κ threshold of 0.60 is adopted as the minimum criterion for acceptable reli-

bility before corpus expansion proceeds. Inter-rater training and annotation protocols will be standardized in advance of the Phase 2 annotation cycle. Following the implementation of the Phase 2 structural detection layer (Section 7.2), a re-evaluation of LLM-based annotation performance against the updated detection criteria is planned; the current $\kappa = 0.211$ baseline serves as the reference point for assessing whether structural pattern detection reduces the judgment load gap between human and LLM annotators.

7.2 Phase 2: Vector Profile Development and Structural Pattern Detection

Phase 2 will investigate whether vector distance measures provide more discriminative power than scalar aggregation for politically framed content. The second-layer detection architecture will target structural manipulation patterns: self-achievement-to-recommendation induction, agreement-seeking sentence-final repetition, numerical chains, scarcity/urgency presentation, and problem-raising-to-self-as-solver. Theoretical basis: Cialdini (1984), Milgram (1974), Janis (1972).

Additionally, EVT variables not currently covered by R8’s lexical architecture—specifically Self-schema manipulation (“discover your true self”) and Personal goal targeting (“what is your dream?”)—represent structural targets for the second detection layer. These patterns correspond to the paragraph-level sequencing identified in Section 2.7 and align with Cialdini’s (1984) commitment and consistency mechanisms. Detecting Self-schema and Personal goal manipulation requires structural pattern recognition beyond the scope of surface lexical density, and is positioned as a Phase 2 design priority.

Phase 2 will also investigate the independent operationalization of scarcity signals as a dedicated detection category (ScarcityRisk). In the current implementation, scarcity-compatible vocabulary is distributed across FearRisk, AbsolutistWords, and ClickbaitRisk without a unified detection architecture. Brock’s (1968) commodity theory and Brehm’s (1966) psychological reactance theory provide the theoretical basis for treating scarcity signaling as a structurally distinct manipulation mechanism warranting independent detection. Five lexical sub-types are proposed for ScarcityRisk operationalization: temporal scarcity (“today only,” “deadline”), quantitative scarcity (“only 3 remaining,” “limited to 50”), informational scarcity (“what only a few know,” “not publicly available”), relational scarcity (“exclusively for you,” “specially selected”), and opportunity scarcity (“this chance will never come again”). Critically, Phase 1 calibration observations indicate that these sub-types rarely appear in isolation in high-risk texts; rather, they

co-occur in combination patterns that produce amplified persuasive effects beyond what linear score aggregation captures (Fromkin & Brock, 1971). Phase 2 will investigate weighted co-occurrence scoring for ScarcityRisk sub-type combinations as an approach to approximating these interaction effects within the current lexical architecture.

7.3 Phase 2: Transformer-Based Contextual Analysis

Following corpus expansion and reliability establishment, integration of transformer-based language models for contextual disambiguation is identified as a significant development direction. This development is framed as complementary rather than replacement.

7.4 Phase 2 Extended: Toward a Non-Linear Detection Model and Multi-Stage Architecture

The current CMI is computed as a linear weighted sum of lexical category scores. This architecture assumes that the persuasive risk contribution of each signal type is additive and independent—an assumption that Phase 1 observations suggest is empirically incomplete.

Two non-linear effects are identified as Phase 2 Extended research priorities. The first is the interaction effect between positive and negative valence signals. It is theoretically hypothesized that texts combining high-density affirmatively valenced vocabulary (EmotionalRisk) with high-density negatively valenced vocabulary (FearRisk) may produce persuasive effects exceeding the prediction of linear score aggregation, consistent with Kahneman’s (2011) negativity dominance finding: negative signals are processed with higher priority than positive signals of equivalent magnitude, so sequential positive-then-negative structures do not cancel but amplify. This interaction hypothesis is proposed as a priority target for Phase 2 empirical investigation with an expanded, independently annotated corpus; the current corpus is insufficient for statistical testing of this effect.

The second is the threshold effect in multi-signal combinations. Phase 1 calibration observations are consistent with the hypothesis that manipulation-compatible signals do not produce proportional effects across their full range, but rather exhibit threshold behavior: below a signal density threshold, effects are sub-linear (weak or absent); above the threshold, effects become supra-linear (rapidly amplifying). This structure is theoretically grounded in Granovetter’s (1978) threshold model of collective behavior, which demon-

strates that individual response thresholds determine when and whether cascades of behavior propagate through a population. Applied to text-based manipulation, the threshold model predicts that: (1) individual readers have personal response thresholds determined by cognitive vulnerability; (2) CMI scores below a reader’s threshold produce minimal compliance effects regardless of absolute score magnitude; and (3) CMI scores that cross the threshold trigger disproportionately amplified responses, consistent with the phase-transition structure of collective behavior cascades.

A candidate mathematical formalization of this threshold structure is the sigmoidal transformation:

$$\text{CMI_nonlinear} = 1 / (1 + e^{-(k(\text{CMI_linear} - \theta)))}$$

where θ represents the individual response threshold and k represents the slope parameter governing the steepness of the transition. This formalization is proposed as a theoretical framework for Phase 3 empirical investigation. The current corpus of 208 documents (valid CMI > 0) with single-rater annotation is insufficient to estimate the parameters θ and k , and the absence of behavioral outcome measures (compliance rates, financial losses, recruitment success) makes empirical validation of the threshold model impossible at this stage. The proposal is presented as a research direction rather than an implemented component.

The multi-stage detection architecture that follows from this theoretical framework is structured as follows. Phase 1 lexical screening (current implementation) functions as a first-stage filter, efficiently excluding documents with CMI below 35 from further processing while directing MEDIUM and HIGH documents —and LOW documents with specific co-occurrence patterns —to second-stage analysis. Phase 2 structural detection applies co-occurrence pattern weighting, paragraph-sequence analysis, and ScarcityRisk combination scoring to the non-cleared subset. Phase 3 contextual and behavioral modeling integrates transformer-based disambiguation, reader vulnerability profiling, and empirical threshold estimation where behavioral outcome data become available. This staged architecture is designed to maximize detection utility at each phase while maintaining transparency about the limitations of each analytical layer.

The threshold model introduced above also provides the conceptual foundation for a complementary construct: the Cognitive Vulnerability Index (CVI). While CMI quantifies the manipulation-compatible signal density of a text—a property of the sender—CVI would quantify the cognitive vulnera-

bility of the reader as the threshold parameter θ that determines when CMI signals produce behavioral effects. The theoretical basis for CVI derives directly from Cialdini’s (1984) empirical observation that susceptibility to consistency-based persuasion increases with age (p. 177), Milgram’s (1974) documentation of authority submission as an individual-difference variable, and Altemeyer’s (1981) Right-Wing Authoritarianism scale, which identifies dispositional tendencies toward authority deference, conventionalism, and out-group hostility as measurable individual differences that correspond structurally to R8’s AuthorityRisk, AbsolutistWords, and EnemyFrame categories. A full CVI operationalization is beyond the scope of the current research; it is proposed here as a Phase 3 theoretical target whose development depends on Phase 2 empirical foundations.

A further theoretical extension concerns the directionality of manipulation vectors. The current R8 architecture treats texts as directed at an external audience—what may be termed an outward manipulation vector. However, organizational and institutional texts often function simultaneously across multiple audience dimensions: externally toward the public or media, internally toward organizational members, and reflexively toward the text’s author as a mechanism of self-justification. Festinger’s (1957) cognitive dissonance theory predicts that inward-directed manipulation—the rhetorical reinforcement of an organization’s own normative framework in the face of contradictory evidence—intensifies rather than resolves when external pressure increases. Empirical candidates for this inward-vector pattern include government crisis-response texts, corporate incident communications, and institutional denial statements, which Phase 1 calibration identified as a systematic false-negative category: high human-label risk but low automated CMI, attributed to the absence of cult- or commercial-genre vocabulary in bureaucratic register. Operationalizing vector directionality—distinguishing outward persuasion from inward self-legitimation—is identified as a Phase 2 corpus design priority and a Phase 3 modeling target.

7.5 Phase 3: Multilingual Extension

Extension to English-language analysis is a potential direction pending Phase 2 validation outcomes.

7.6 Phase 3: AI-Generated Text Analysis

A longer-term research direction extends R8’s framework to the analysis of AI-generated communication. Whether R8’s lexical categories are sensitive

to sycophantic bias patterns in AI-generated text is an open empirical question that cannot be meaningfully investigated until the framework’s validity on human-authored text has been established in Phase 2.

7.7 Phase 3: Multimodal Extension

Extension to audio and video content is technically feasible by placing a speech-to-text conversion stage (e.g., Whisper) prior to the R8 pipeline. Theoretical grounding: Janis (1972) documented that defective decision-making patterns manifest in leaders’ public statements, providing a basis for extending manipulation signal detection to spoken discourse.

8. Conclusion

R8 presents a first-generation lexical framework for approximating cognitive manipulation risk in Japanese-language text. By operationalizing 12 theoretically grounded risk categories into a single composite index —the Cognitive Manipulation Index (CMI) —the system provides a transparent, reproducible, and scalable screening tool for identifying texts that warrant closer scrutiny. The system is explicitly positioned not as a definitive classifier of manipulative intent, but as an initial flag that complements rather than replaces human judgment.

The theoretical framing of R8 integrates cognitive psychology (Kahneman, 2011), social psychology (Janis, 1972; Milgram, 1974; Zimbardo, 2007), and Japanese organizational sociology (Tobe et al., 1991; Yamamoto, 1977; Nakane, 1967), drawing additionally on clinical psychology accounts of sub-threshold manipulation (Symington, 1993). The novel contribution of this research lies in integrating these previously disconnected theoretical domains under the unified lens of “linguistic signals targeting System 1 cognitive processing,” and operationalizing this integration as a quantifiable lexical instrument for Japanese text.

Phase 1 calibration identified three principal findings. First, CMI achieves its highest detection performance for politically framed content combining EnemyFrame, Propaganda, and FearRisk simultaneously. Second, commercial manipulation content systematically evades HIGH classification through surface-positive vocabulary, establishing the central diagnostic finding of Phase 1. Third, calibration confirmed that lexical density scoring

produces systematic false negatives for sophisticated manipulation and systematic false positives for argumentative analytical prose, establishing the empirical ceiling of the lexical-only approach.

The limitations of the current implementation are substantial and explicitly acknowledged. These limitations define the Phase 2 research agenda: validated corpus expansion with independent inter-rater annotation; development of a second detection layer targeting structural manipulation patterns; and construction of a contrastive corpus enabling false-negative detection.

R8 is released as open-source software with the explicit acknowledgment that it is a first approximation. The value of this release lies in establishing a transparent, reproducible baseline—one that documents the ceiling of lexical detection before the transition to structural and semantic analysis, and that positions Japanese-language cognitive manipulation research within the broader international conversation on computational detection of persuasive and deceptive text.

References

- Altemeyer, B. (1981). *Right-wing authoritarianism*. University of Manitoba Press.
- Arai, N. (2018). *Robotto wa toudai ni haireruka* [Can a robot get into Tokyo University?] (Rev. ed.). Shin-yo-sha.
- Baumeister, R. F., & Leary, M. R. (1995). The need to belong: Desire for interpersonal attachments as a fundamental human motivation. *Psychological Bulletin*, 117(3), 497–529.
- Brehm, J. W. (1966). *A theory of psychological reactance*. Academic Press.
- Brock, T. C. (1968). Implications of commodity theory for value change. In A. G. Greenwald, T. C. Brock, & T. M. Ostrom (Eds.), *Psychological foundations of attitudes* (pp. 243–75). Academic Press.
- Cialdini, R. B. (1984). *Influence: The psychology of persuasion*. Harper Business.
- Eccles, J. S., & Wigfield, A. (2002). Motivational beliefs, values, and goals. *Annual Review of Psychology*, 53, 109–32.

- Festinger, L. (1957). *A theory of cognitive dissonance*. Stanford University Press.
- Fromkin, H. L., & Brock, T. C. (1971). A commodity theory analysis of persuasion. *Public Opinion Quarterly*, 35(3), 367–77.
- Granovetter, M. (1978). Threshold models of collective behavior. *American Journal of Sociology*, 83(6), 1420–443.
- Granovetter, M., & Soong, R. (1983). Threshold models of diffusion and collective behavior. *Journal of Mathematical Sociology*, 9(3), 165–79.
- Hathaway, S. R., & McKinley, J. C. (1943). *The Minnesota Multiphasic Personality Inventory*. University of Minnesota Press.
- Janis, I. L. (1972). *Victims of groupthink: A psychological study of foreign-policy decisions and fiascos*. Houghton Mifflin.
- Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux.
- Milgram, S. (1963). Behavioral study of obedience. *Journal of Abnormal and Social Psychology*, 67(4), 371–78.
- Milgram, S. (1974). *Obedience to authority: An experimental view*. Harper & Row.
- Nakane, C. (1967). *Tate-shakai no ningen kankei* [Human relations in a vertical society]. Kodansha.
- Ross, R. A. (2014). *Cults inside out: How people get in and can get out*. CreateSpace.
- Sunstein, C. R. (2017). *#Republic: Divided democracy in the age of social media*. Princeton University Press.
- Symington, N. (1993). *Narcissism: A new theory*. Karnac Books.
- Tobe, R., Teramoto, Y., Kamata, S., Sugino, Y., Murai, T., & Nonaka, I. (1991). *Shippai no honshitsu* [The essence of failure]. Diamond.
- Yamamoto, S. (1977). *Kuuki no kenkyuu* [A study of “air”]. Bungeishunju.
- Zimbardo, P. G. (2007). *The Lucifer effect: Understanding how good people turn evil*. Random House.
- Anthropic. (2026). Claude Sonnet 4.6 and Claude Opus 4.6 [Large language model]. <https://www.anthropic.com>

- Ben-Ghiat, R. (2020). *Strongmen: Mussolini to the present*. W. W. Norton.
- Da San Martino, G., Yu, S., Barrón-Cedeño, A., Petrov, R., & Nakov, P. (2019). Fine-grained analysis of propaganda in news articles. *EMNLP 2019*, 5636–646.
- Da San Martino, G., Barrón-Cedeño, A., Wachsmuth, H., Petrov, R., & Nakov, P. (2020). SemEval-2020 task 11: Detection of propaganda techniques in news articles. *SemEval 2020*, 1377–414.
- Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159–74.
- Pennebaker, J. W., Boyd, R. L., Jordan, K., & Blackburn, K. (2015). The development and psychometric properties of LIWC2015. University of Texas at Austin.
- Wang, Y., Yang, I., Hassanpour, S., & Vosoughi, S. (2024). MentalManip: A dataset for fine-grained analysis of mental manipulation in conversations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 3747–764). Association for Computational Linguistics.
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